An interior cut cannot first share the exceptional crossing

Human Compression project: L044 continuation

Retain the full critical six-point-head hypotheses and their paid predecessor proofs. In particular $G \le \mathrm{Alt}(\Omega)$ is transitive, $|\Omega| = n = 2r \ge 18$, its distinguished involutions form a connected string C-marking, and all maximal omissions are intransitive. The actual first three generators form the supported six-point head on D. The critical tail is the actual natural alternating group on $(\Omega \setminus D) \cup \{6\}$, with the notation of *Pure alternating kernels in the critical prefixes.* Write $P_k = G_{\le k}$ and let C_k be its orbit containing D. We retain the entire proof of that note, including

$$\mathrm{Alt}(C_k) \le P_k, \qquad P_k = \{(s,u) \in \mathrm{Sym}(C_k) \times J_k : \operatorname{sgn} s = \operatorname{sgn} u\}, \tag{1}$$

where J_k is the literal image on $\Omega \setminus C_k$. All eligible edges below refer to full ambient omissions.

Lemma 1 (The two sides of a later unique cut)**.** *Suppose $j \ge 5$ has exactly one eligible edge. Its two omission orbits are $C = C_{j-1}$ and $A = \Omega \setminus C$. Every generator of index greater than j fixes C pointwise. Consequently every shared label at this cut is smaller than j; if there is a first shared label, it is at least 3.*

Proof. Put $P = G_{<j}$ and $Q = G_{>j}$. These groups commute. By (1), P contains the actual natural $\mathrm{Alt}(C_{j-1})$, on at least eight points. Every element of Q therefore preserves that unique nontrivial orbit and fixes it pointwise: a permutation centralizing a natural alternating group of degree at least four is trivial on its support. Thus the $G_j = PQ$ orbit containing D is exactly C_{j-1}. A unique eligible edge and transitivity of G imply that G_j has exactly two orbits, since adding that one cross-orbit edge must connect all of them. This proves the orbit assertion. The head fixes $\Omega \setminus D$, so its three labels cannot be shared across this cut. The fixation by Q excludes every label greater than j as well. □

Theorem 2 (Exclusion of the interior first-shared case)**.** *Suppose $5 \le j < r-1$ has exactly one eligible edge. Then its first shared label cannot be 3.*

Proof. Assume the first shared label is 3. Reverse the entire string, writing $s_t = \rho_{r-1-t}$, and put

$$i = r-1-j \ge 1, \qquad h = r-4, \qquad d = h-i = j-3 \ge 2.$$

In this reversed order the unique crossing has index i, and the last shared label is h. Its earlier subgroup $L = \langle s_t : t < i \rangle = G_{>j}$ fixes C pointwise by Lemma 1. The tail past h is precisely the original supported head on D. It fixes A, and the size of its fixed set is therefore exactly

$$F = n - 6 = 2h + 2. \tag{2}$$

If L is transitive on A, the complete transitive-side support lemma gives $h = i+1$, contrary to $d \ge 2$. We may consequently use the complete intransitive-side orbit and support lemmas. Their hypotheses hold literally: i is interior; the second orbit C contains D and is nontrivial; L fixes C; and the tail past the last shared label fixes A. Let k be the common size of the faithful L-orbits in A. Those lemmas give $k \ge i+1$ and the following bounds, with their full proofs retained.

For $d = 2$, if s_h exchanges the first active L-orbit with another one, then

$$F \ge 3k + 2 \ge 3i + 5.$$

Together with (2), this forces $i \le 1$, and hence $r = h + 4 = i + 6 \le 7$, contrary to $r \ge 9$. If that active orbit is preserved nontrivially instead, the complete internal-orbit lemma gives $F \ge 2h+3$, directly contradicting (2). These are the two cases of the paid distance-to-an-active-orbit argument when $d = 2$.

For $d \ge 3$, that argument and the fixed-prefix path bound give

$$F \ge |A| + d \ge dk + d \ge d(i + 2).$$

Equation (2) becomes $i(d-2) \le 2$. Since $i \ge 1$ and $r = i + d + 4 \ge 9$, its only possibilities are

$$r = 9, \quad (i, d) = (2, 3) \quad \text{or} \quad (i, d) = (1, 4). \tag{3}$$

Both have $F = 12$, and all the preceding inequalities are equalities. In the first case $k = 3$ and A consists of three L-orbits of size three. The two independent involutions generating L act faithfully on each orbit; each restriction is thus a transposition. Since L fixes C, each generator is a product of three transpositions on Ω, contradicting $G \le \mathrm{Alt}(\Omega)$.

It remains to exclude $(i, d) = (1, 4)$. Here $j = 7$, $|A| = 8$, and $L = \langle \rho_8 \rangle$ has four two-point orbits in A. Write $\lambda = \rho_8$: it flips each of these four pairs and fixes C. The complete orbit-distance proof starts at the pair containing the crossing endpoint and reaches a pair on which $s_h = \rho_3$ is nontrivial, using at least $d - 1 = 3$ intervening labels. There are exactly four pairs. A shortest such path therefore visits all four, and ρ_3 fixes the first three pointwise. It cannot exchange the last pair with another pair, because those earlier pairs are fixed. It follows that the restriction $\rho_3|_A$ is exactly the transposition on the last pair. This conclusion uses the full active-orbit argument, not merely its numerical inequality.

Now take $P = G_{<7} = P_6$ and let J be its image on A. Every element of P commutes with λ, so J preserves the four pairs. The group $G_7 = P\langle\lambda\rangle$ is transitive on A, whereas λ fixes each pair setwise. Hence J is transitive on the set of four pairs. It contains the single pair flip $\rho_3|_A$. Conjugating that flip by J supplies the flip on every pair; their product is $\lambda|_A$. This product is even, being a product of four disjoint transpositions. Formula (1), with $C = C_6$ and $J = J_6$, now puts the literal permutation $(1_C, \lambda|_A) = \rho_8$ in P_6. But actual ambient IP gives $P_6 \cap \langle \rho_8 \rangle = 1$, since the distinguished index sets are disjoint. This is the final contradiction in (3), proving the theorem. □

Attribution and exact remaining scope. The source route is the exceptional exit from the reversed transitive-tail argument in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Propositions 5.26 and 5.28, physical pages 34 and 36–37. The complete transitive-side, intransitive-side, active-orbit, internal-orbit and fixed-prefix proofs, with every classical and finite input, remain paid. The near-equality count and the final extraction of the even four-pair flip are the new argument here. The prefix-kernel proof attributes the initial parity-fibre suggestion to the root collaborator.

The endpoint $j = r - 1$ is excluded from Theorem 2; it has $i = 0$ after reversal, so the interior orbit bounds do not apply. Other first shared indices, the opposite heads, the terminal singleton and complete L044 recovery remain separate. No computational, formal, whole-original, conjecture or numerical-compression credit is claimed.